

Science—Physics: How Light Works

Subject: Science—Physics

Light is a form of electromagnetic radiation that transfers energy through electric and magnetic fields. In a vacuum, light travels at approximately 3.0×10^8 meters per second. Unlike sound, light does not need a material medium, so it can travel through empty space from the Sun to Earth. Visible light is only a small part of the electromagnetic spectrum. Radio waves, microwaves, infrared, ultraviolet, X-rays, and gamma rays have different wavelengths and frequencies, but they are all electromagnetic waves.

Wavelength is the distance between repeating points on a wave, while frequency is the number of cycles per second. For electromagnetic waves, speed, wavelength, and frequency are related by $c = f\lambda$. When frequency increases, wavelength decreases if speed remains constant. The energy of a photon is related to frequency by $E = hf$, where h is Planck's constant. This means higher-frequency radiation, such as ultraviolet or X-rays, carries more energy per photon than lower-frequency radio waves. Excess exposure to high-energy radiation can damage living tissue, so shielding and controlled exposure are important.

Light can be reflected, refracted, absorbed, transmitted, scattered, or emitted. Reflection occurs when light bounces from a surface. A smooth surface can produce a clear image, while a rough surface scatters light in many directions. Refraction is the bending of light when it enters a material where its speed changes. A pencil may appear bent in water because light from the submerged part changes direction as it leaves the water. Lenses use refraction to converge or diverge light. A convex lens can bring parallel rays toward a focus, while a concave lens spreads them outward.

White light can be separated into colors because different wavelengths refract by different amounts. A prism produces a spectrum, and raindrops can create rainbows by refraction, internal reflection, and dispersion. Objects appear colored because they absorb some wavelengths and reflect or transmit others. A green leaf usually reflects more green light than other visible wavelengths. Dark surfaces absorb more light energy and may become warmer. Materials that allow much light through are transparent, while translucent materials scatter light and opaque materials block or absorb it.

Light behaves both as a wave and as particles called photons. Wave behavior explains interference, diffraction, and polarization. The photoelectric effect shows that light can transfer energy in discrete packets to electrons. Modern technologies use both descriptions. Fiber-optic cables guide light through repeated total internal reflection and carry digital information over long distances. Cameras, microscopes, telescopes, solar cells, lasers, medical imaging devices, and optical sensors all depend on controlled interactions between light and matter. Understanding reflection, refraction, wavelength, frequency, and photon energy helps explain both everyday observations and advanced instruments.